Overfitting:
Under fitting

Feature selection
Multicollinearity

Overfitting:

If training Data has Very Good Accuracy while Testing Data has Bad Accuracy then this condition is called overfitting.

To solve such a problem need to perform **hyper-parameter tuning, OR Regularization .**

Under-fitting :

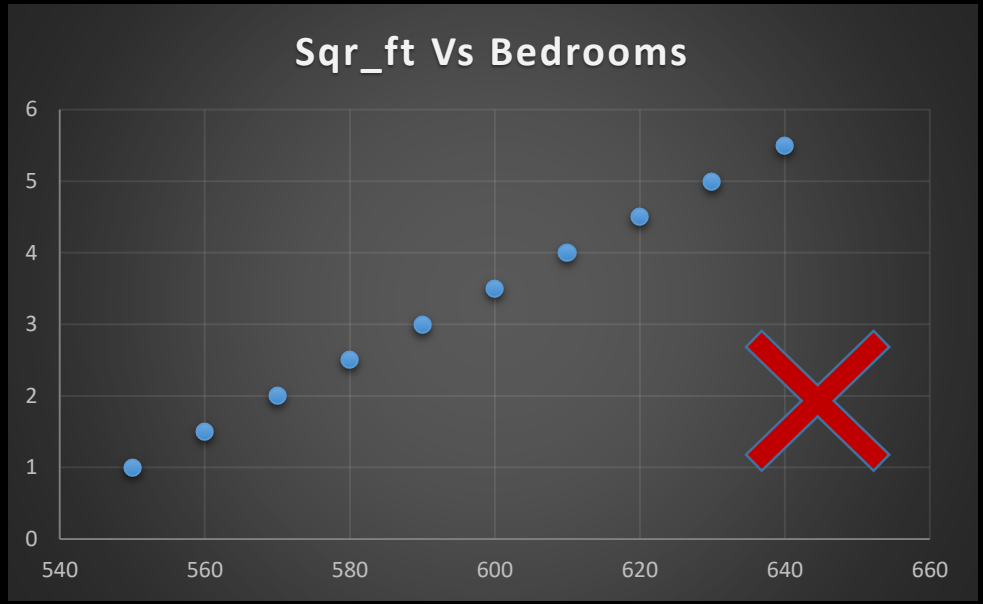
If training Data has Very Bad Accuracy, while Testing Data has Bad or Good Accuracy then this condition is called Under fitting.

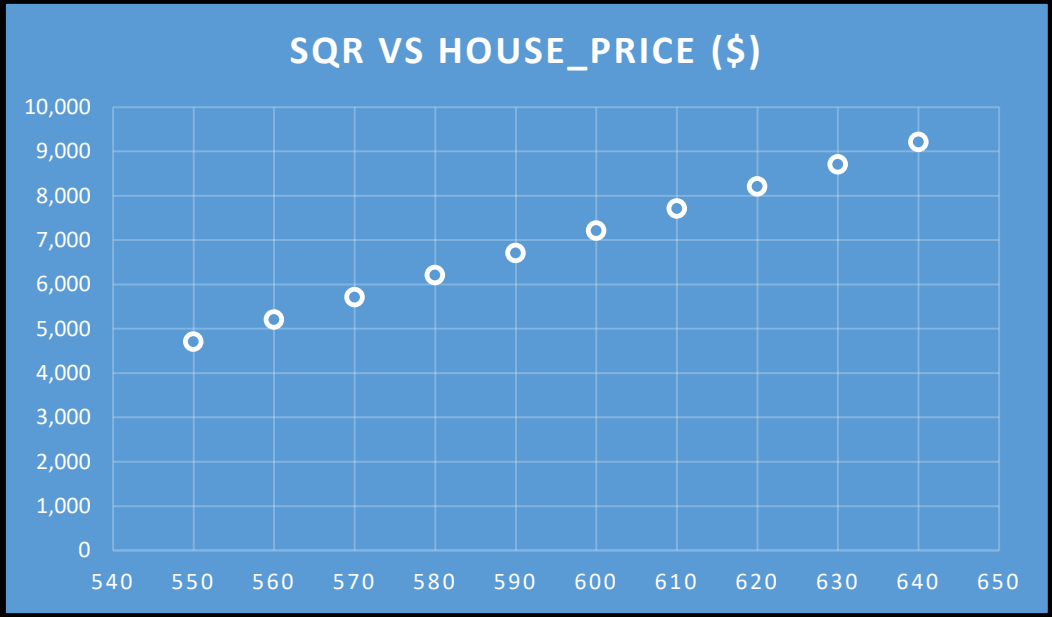
To solve such problem we need to do the retraining of the model.

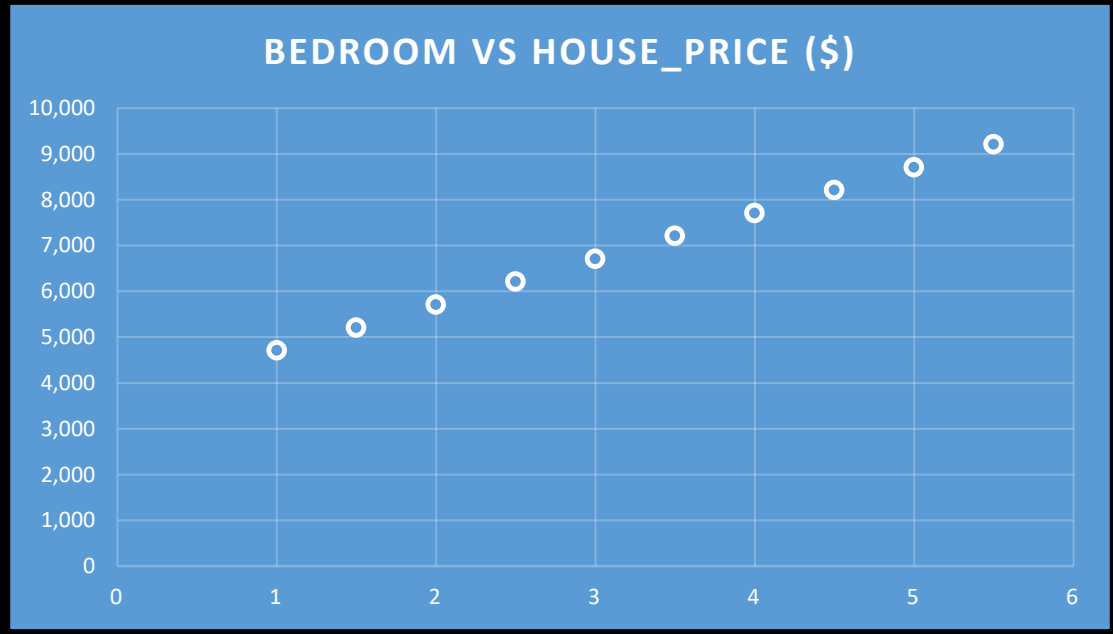
Feature Selection: The process of choosing the most relevant, non-redundant features to simplify models, speed up training.

Multicollinearity:

Sqr Ft	Bedrooms	Bathrooms	Age_Years	House Price ($)
550	1	1	21	4,712
560	1.5	1	15	5,212
570	2	2	2	5,712
580	2.5	2	28	6,212
590	3	3	14	6,712
600	3.5	3	26	7,212
610	4	4	7	7,712
620	4.5	4	18	8,212
630	5	5	11	8,712
640	5.5	5	5	9,212







Multicollinearity:
A situation where **two or more independent features** in a dataset are strongly correlated.
This may causing model instability and unreliability in model training.

Multicollinearity is a situation in regression analysis where **two or more independent variables are highly correlated with each other**. This means they contain similar information about the dependent variable, making it difficult for the model to distinguish the individual effect of each predictor.

Square_Feet	Bedrooms	Bathrooms	Age_Years	House Price ($)
550	1	1	21	4,712
560	1.5	1	15	5,212
570	2	2	2	5,712
580	2.5	2	28	6,212
590	3	3	14	6,712
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630	5	5	11	8,712
640	5.5	5	5	9,212